Lesson notes · for you

Make it light up

IoT · Level 2 · Lesson 1

Materials
Two AA batteries, tape, and kitchen foil or three short wires; A torch bulb, or a red LED with a 100 Ω resistor; Ten everyday things to test: coins, keys, a pencil, a ruler, paper, foil; No kit? A phone or computer for the circuit lab

Learning goals

- Build a complete circuit with a battery, wires and a bulb or LED.
- Sort materials into conductors and insulators by testing them.
- Explain why an LED needs a resistor and why a short circuit is dangerous.

Activity: Conductor hunt

1. Tape two AA batteries end to end, + to -. Connect the bulb (or the LED and its 100 Ω resistor) with foil strips or wires, and leave a gap between two free ends.
2. Touch the two free ends together for a moment. The bulb should light. If it doesn't, check the bulb's two contacts and the LED's long leg.
3. Write ten everyday things in a table, and predict for each one: lights, or stays dark.
4. Touch both free ends to each thing, and write the result next to your prediction.
5. Sort the ten into conductors and insulators. Circle any you got wrong, and write why a wire is copper inside plastic.

Batteries only, never a wall socket. Ask an adult to cut or strip wires. Never join the battery's + and - with a bare wire or foil: that's a short circuit. Keep button batteries away from small children.

Discussion

- Which result surprised you, and why?
- Why does an LED need a resistor when a torch bulb doesn't?
- Why is a wire copper inside plastic, and not plastic inside copper?

Check yourself

- Name one conductor, one insulator, and one safety rule for circuits.

Try it this week
Conductor hunt: Make a tester from two AA batteries, a torch bulb (or a red LED with a 100 Ω resistor) and strips of kitchen foil or wire, leaving a gap between two free ends. Predict first, then test ten things at home or at school, and sort them into conductors and insulators in a table. Finish with one surprise: something you predicted wrong. No batteries? Write your predictions now, test them with your school's kit, and use the circuit lab for the LED.

Worksheet

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Name _____ Class _____ Date _____

1 **Musa wants to test his battery, so he holds a bare wire from its + end straight to its – end. There's no bulb in the loop. What happens?**

- ◯ Nothing: with no bulb, no current flows
- ◯ The wire and the battery quickly get hot
- ◯ The battery goes flat slowly, but safely

2 **You touch each of these across the gap in your bulb tester. Which will light the bulb?**

- ☐ A steel paper clip
- ☐ A coin wrapped in sticky tape
- ☐ The metal zip on your bag
- ☐ A dry leaf
- ☐ A plastic ruler
- ☐ A strip of kitchen foil

3 **If an LED stays dark in your circuit, it must be broken.**

- ◯ True
- ◯ False

4 **In the circuit lab, a blue LED on the 3 V battery stays dark, even with no resistor at all. Why?**

- ◯ A blue LED takes about 3 V, so none is left to push a current
- ◯ Blue LEDs need far more current than a 3 V battery can give
- ◯ With no resistor, it burnt out the moment it was connected

5 **A red LED on a 9 V battery, with a 470 Ω resistor. Work it out.**

The LED takes 2 V, so _____ V is left for the resistor. The current is about _____ mA, which is safe.

Word bank: 19 · 7 · 9 · 15

✎ **Where at home could a short circuit happen by accident? How would you stop it?**

Check your answers

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1. **The wire and the battery quickly get hot**

   With nothing in the loop to limit it, a huge current flows: that's a short circuit.

2. **A steel paper clip; A strip of kitchen foil; The metal zip on your bag**

   The paper clip, the foil and the zip are bare metal, so they close the loop.

3. **False**

   Check the causes that cost nothing first.

4. **A blue LED takes about 3 V, so none is left to push a current**

   An LED always takes its own voltage first: about 2 V for red, about 3 V for blue.

5. **The LED takes 2 V, so 7 V is left for the resistor. The current is about 15 mA, which is safe.**

   9 − 2 = 7 V for the resistor, and 7 ÷ 470 ≈ 0.015 A, about 15 mA: under 20 mA, so it's safe.